HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2013)

Student No.: 01 Name: _____ Marks: / 54

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The solution of $x - \frac{x-1}{2} > 5$ or $1 < x - 11$ is

   A. $x > 9$

   B. $x > 10$

   C. $x > 11$

   D. $x > 12$

(DSE-2013 PII Q05)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

2. Let k be a constant. Solve $(x-k)^2 = 4k^2$.

A. $x = 3k$

B. $x = 5k$

C. $x = -k$ or $x = 3k$

D. $x = -3k$ or $x = 5k$

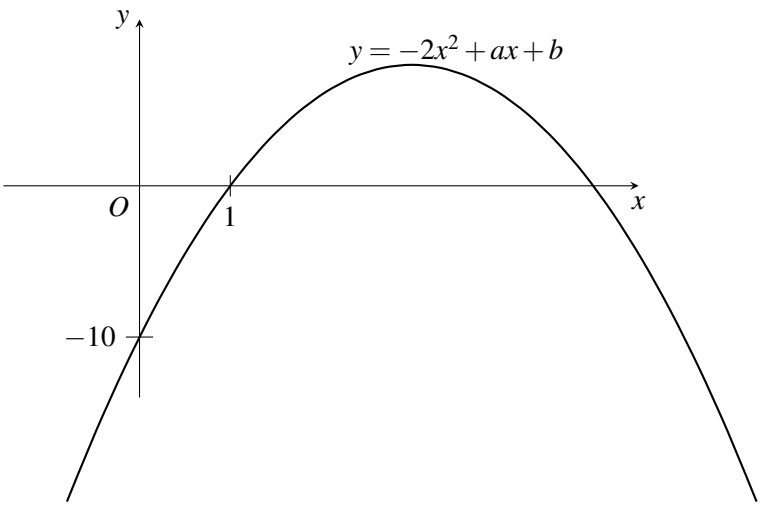
(DSE-2013 PII Q06)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

3. The figure shows the graph of $y = -2x^2 + ax + b$. The equation of its axis of symmetry is



A. $x = 2$

B. $x = 3$

C. $x = 5$

D. $y = 8$

(DSE-2013 PII Q07)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

4. Susan sells two cars for $80 080 each. She gains 30% on one and loses 30% on the other. After the transactions, Susan

   A. loses $15 840.

   B. gains $5 544.

   C. gains $10 296.

   D. has no gain and no loss.

(DSE-2013 PII Q10)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

5. A sum of $50 000 is deposited at 8% per annum for 1 year, compounded monthly. Find the interest to the nearest dollar.

A. $4 000

B. $4 122

C. $4 143

D. $4 150

(DSE-2013 PII Q11)

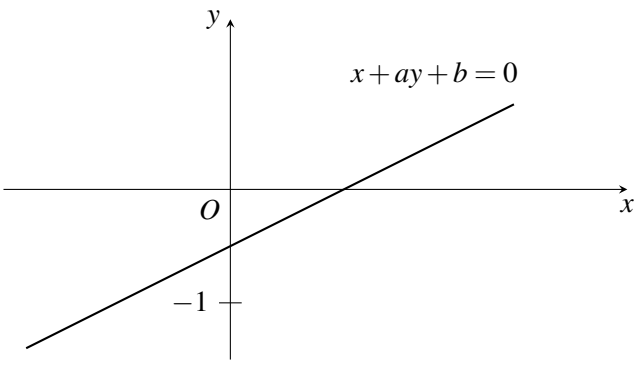
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

6. The figure shows $x+ay+b=0$. Which statements are true?

I. $a<0$ II. $b<0$ III. $a<b$



A. I and II only

B. I and III only

C. II and III only

D. I, II and III

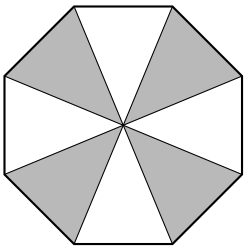
(DSE-2013 PII Q14)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

7. In the figure, the regular octagon is divided into eight identical isosceles triangles and four of them are shaded. The number of axes of reflectional symmetry of the octagon is



A. 2

B. 4

C. 8

D. 16

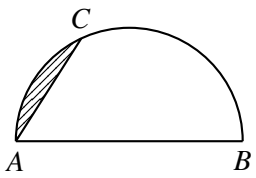
(DSE-2013 PII Q15)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

8. In the figure, the diameter of the semicircle ABC is 3 cm. If chord $AC = 2$ cm, find the area of the shaded region corrected to the nearest $0.01\,\text{cm}^2$.



A. $0.23\,\text{cm}^2$

B. $0.52\,\text{cm}^2$

C. $0.64\,\text{cm}^2$

D. $1.07\,\text{cm}^2$

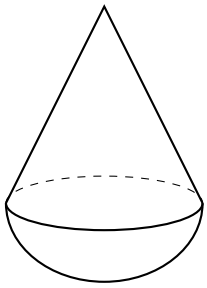
(DSE-2013 PII Q16)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

9. In the figure, the solid consists of a right circular cone and a hemisphere with a common base. The base radius and the height of the circular cone are 3 cm and 4 cm respectively. Find the total surface area of the solid.



A. $30\pi\,\text{cm}^2$

B. $33\pi\,\text{cm}^2$

C. $48\pi\,\text{cm}^2$

D. $51\pi\,\text{cm}^2$

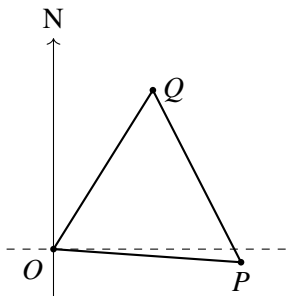
(DSE-2013 PII Q17)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

10. In the figure, the bearing of P from O is $S86°E$ and the bearing of from O is $N32°E$. If P and Q are equidistant from O, then the bearing of P from Q is



A. N24°W

B. N27°W

C. S24°E

D. S27°E

(DSE-2013 PII Q20)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

11. If $0^\circ < x < 90^\circ$, which must be true?

I. $\tan x \tan(90^\circ - x) = 1$

II. $\sin x - \sin(90^\circ - x) < 0$

III. $\cos x + \cos(90^\circ - x) > 0$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2013 PII Q23)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

12. The equation of the circle C is $2x^2+2y^2-4x+8y-5=0$. The coordinates of the points P and Q are $(-1,2)$ and $(4,0)$ respectively. Which of the following is/are true?

I. The radius of C is 5.

II. The midpoint of PQ lies outside C.

III. If G is the centre of C, then $\angle PGQ$ is acute angle.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2013 PII Q25)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

13. Two numbers are randomly drawn at the same time from seven cards numbered 1, 2, 3, 4, 5, 6 and 7 respectively. Find the probability that the product of the numbers drawn is an odd number.

A. $\frac{2}{7}$

B. $\frac{4}{7}$

C. $\frac{12}{49}$

D. $\frac{16}{49}$

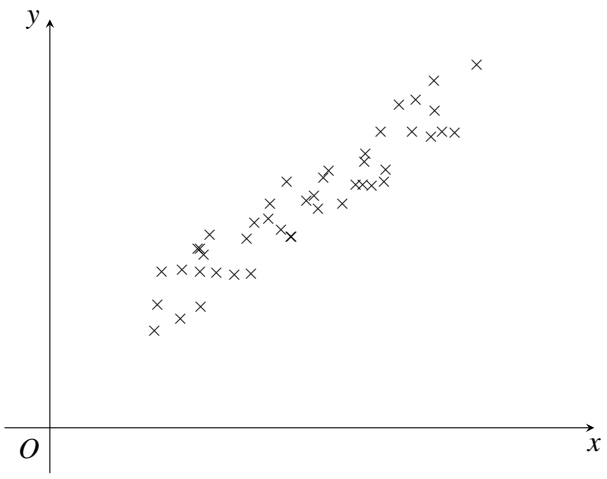
(DSE-2013 PII Q26)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

14. The scatter diagram shows the relation between x and y. Which may represent the relation?



A. y increases when x increases.

B. y decreases when x increases.

C. y varies inversely as x^2.

D. y varies directly as x^{-3}.

(DSE-2013 PII Q28)

Answers written in the margins will not be marked.

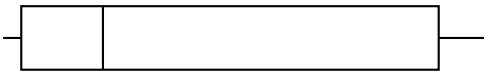
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

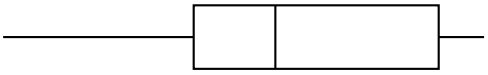
15. The stem-and-leaf diagram shows hourly wages. Which box-and-whisker diagram may represent them?

4	0 2 2 2 4 4 4 7
5	0 0 1 2 2 6 8 9
6	3 5 5 7
7	0
8	2 6
9	5

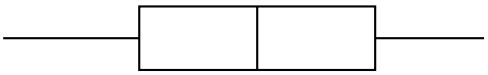
A.



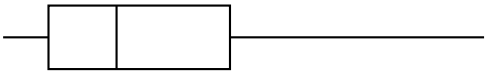
B.



C.



D.



(DSE-2013 PII Q29)

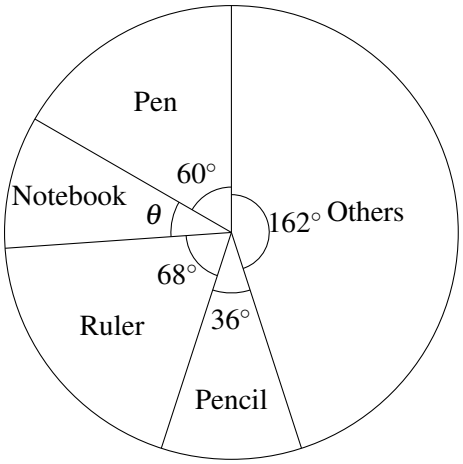
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

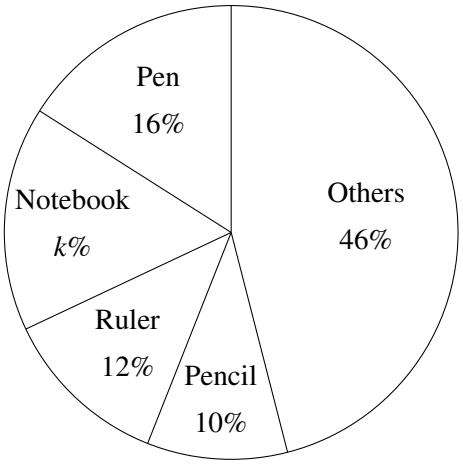
Answers written in the margins will not be marked.

16. The pie charts below show the distributions of the profits of stationery shop$X$ and stationery shop Y from the sales of stationery in a certain month. Which of the following must be true?

Distribution of the profits of stationery shop X



Distribution of the profits of stationery shop Y



A. In that month, the profit from the sales of pencils of stationery shop X is the same as that of stationery shop Y.

B. In that month, the total profit from the sales of pens and notebooks of stationery shop X is less than the total profit from the sales of rulers and pencils of the shop.

C. $k = 14$

D. $\theta = 36^\circ$

(DSE-2013 PII Q30)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

17. The L.C.M. of a^2+4a+4, a^2-4 and a^3+8 is

A. $a+2$

B. $(a-2)(a+2)^2(a^2-2a+4)$

C. $(a-2)(a+2)^2(a^2+2a+4)$

D. $(a-2)(a+2)^4(a^2-2a+4)$

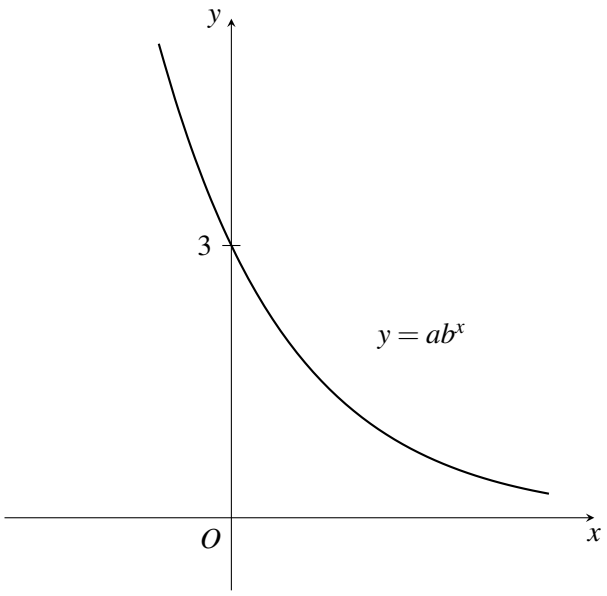
(DSE-2013 PII Q31)

Answers written in the margins will not be marked.

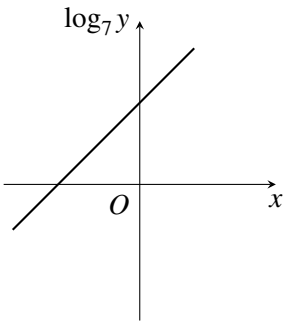
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

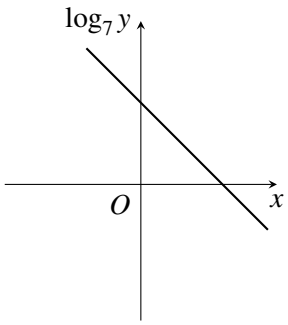
18. The figure below shows the graph of $y = ab^x$, where a and b are constants. Which of the following graphs may represent the relation between x and $\log_7 y$?



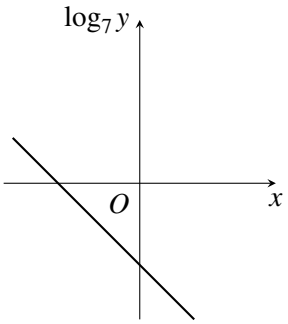
A.



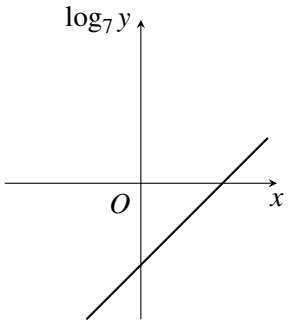
B.



C.



D.



(DSE-2013 PII Q32)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

19. $A00000E00001_{16} =$

A. $10 \times 16^{11} + 14 \times 16^5 + 17$

B. $11 \times 16^{11} + 15 \times 16^5 + 17$

C. $10 \times 16^{12} + 14 \times 16^6 + 272$

D. $11 \times 16^{12} + 15 \times 16^6 + 272$

(DSE-2013 PII Q33)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

20. If $x - \log y = x^2 - \log y^2 - 10 = 2$, then $y =$

A. 100

B. 2 or -4

C. $\frac{1}{100}$ or 10000

D. $\frac{1}{10000}$ or 100

(DSE-2013 PII Q34)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

21. If $\alpha \neq \beta$ and $\begin{cases} 3\alpha = \alpha^2 - 5 \\ 3\beta = \beta^2 - 5 \end{cases}$, then $\alpha\beta =$

A. 3

B. -3

C. 5

D. -5

(DSE-2013 PII Q35)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

22. The real part of $i+2i^2+3i^3+4i^4$ is

A. 2

B. -2

C. 6

D. -6

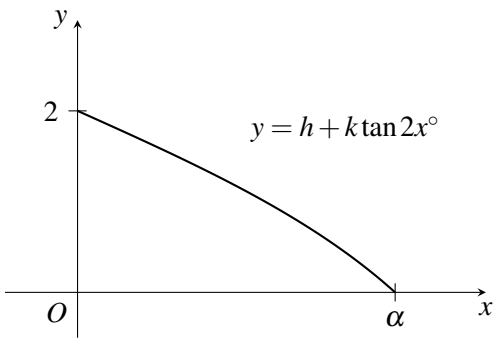
(DSE-2013 PII Q36)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

23. Let h and k be constants. The figure shows the graph of $y = h + k\tan 2x^\circ$, where $0 \le x \le \alpha$. Which of the following are true?



I. $h > 0$

II. $k < 0$

III. $\tan\alpha = \frac{1}{k}$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2013 PII Q39)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

24. If the height of a regular tetrahedron has height 2 cm. then the volume of the tetrahedron is

A. $2\,\text{cm}^3$

B. $\sqrt{3}\,\text{cm}^3$

C. $\sqrt{6}\,\text{cm}^3$

D. $3\sqrt{3}\,\text{cm}^3$

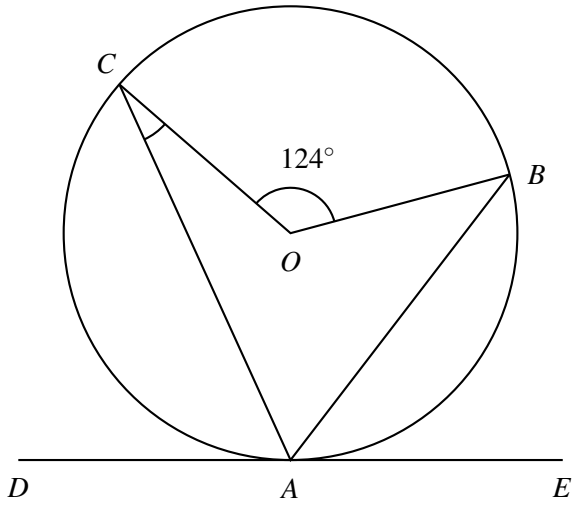
(DSE-2013 PII Q40)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

25. In the figure, O is the centre of the circle ABC, and DE is tangent to the circle at A. If AB is the angle bisector of $\angle CAE$, then $\angle ACO =$



A. 26°

B. 28°

C. 31°

D. 34°

(DSE-2013 PII Q41)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

26. Find the range of k such that the circle $x^2 + y^2 + 2x - 2y - 7 = 0$ and the straight line $3x - 4y + k = 0$ intersect.

A. $-8 < k < 22$

B. $-8 \leq k \leq 22$

C. $k < -22$ or $k > 8$

D. $k \leq -22$ or $k \geq 8$

(DSE-2013 PII Q42)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

27. Let O be the origin. If the coordinates of the points A and B are $(0, 12)$ and$(30, 12)$ respectively, then the $y-$coordinate of the circumcentre of ΔOAB is

A. 6.

B. 8.

C. 12.

D. 15.

(DSE-2013 PII Q43)

I can finish _____ questions

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.